

Total No. of Questions : 10]

SEAT No. :

P2056

[Total No. of Pages : 2

[5059]-661

B.E. (Information Technology)
INFORMATION AND CYBER SECURITY
(2012 Pattern) (Semester - I)

Time : 2.30 Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) *Answers Question 1 or 2, 3 or 4, 5 or 6, 7 or 8 and 9 or 10.*
- 2) *Neat diagrams must be drawn whenever necessary.*
- 3) *Figures to the right indicate full marks.*
- 4) *Assume suitable data, if necessary.*

- Q1)** a) List and briefly define types of cryptanalytic attacks based on what is known to the attacker. [6]
- b) Determine gcd (24140, 16762). [4]

OR

- Q2)** a) Using the extended Euclidean algorithm, find the multiplicative inverse of [6]
- i) 1234 mod 4321
- ii) 24140 mod 40902
- b) What is the difference between a monoalphabetic cipher and a polyalphabetic cipher? [4]

- Q3)** a) What characteristics are needed in a secure hash function? [6]
- b) In what order should the signature function and the confidentiality function be applied to a message, and why? [4]

OR

- Q4)** a) What are the properties a digital signature should have? [6]
- b) What four requirements were defined for Kerberos? [4]

P.T.O.

Q5) a) Give examples of applications of IPSec. What services are provided by IPSec? What is the difference between transport mode and tunnel mode in IPSec? [8]

b) What protocols comprise SSL. Draw a neat diagram? What is the difference between an SSL connection and an SSL session? [8]

OR

Q6) a) What services are provided by the SSL Record Protocol? What steps are involved in the SSL Record Protocol transmission? [8]

b) What are three benefits that can be provided by an intrusion detection system? What is the difference between statistical anomaly detection and rule-based intrusion detection? [8]

Q7) a) What is cybersquatting. Who are cybersquatters and how does it work. [8]

b) What are social engineering attacks and classify and explain them? [8]

OR

Q8) a) What is cyberstalking. Explain cyberstalking and explain how it works. [8]

b) Classify and explain cybercrimes against property. [8]

Q9) Write notes on : [18]

a) Viruses

b) Worms and logic bombs

c) Botnets

OR

Q10 Write notes on : [18]

a) Cloud computing and cybercrimes

b) Indian legal perspective on cybercrimes

c) Software Piracy

